Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

EIN 4360 Facilities Planning and Material Handling

Chapter 11: Evaluating & Selecting the Facilities Plan

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POWERING THE NEW ENGINEER TO TRANSFORM THE FUTURE

Summary

- We will discuss how to select the best facility plan from our developed alternatives
- Looking outside of just the objective functions
- Two methods for evaluating and selecting alternatives
  - *Ranking*
  - *Analytical Hierarchy Process (AHP)*

Evaluating Facilities Plans

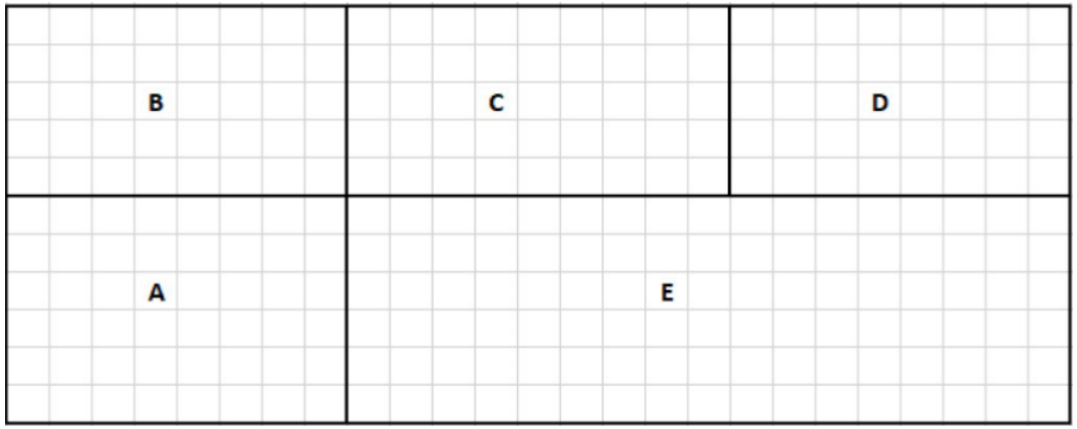
- **List the positive & negative** aspects of each alternative
- **Rank the performance** of each alternative against the criteria
- Perform a **weighted factor comparison** of the alternatives
- Determine the **economic performance** of each alternative over a specified planning horizon

Rank the Performance: Example

- You are evaluating 3 possible locations for a new machine tool storage: (3,4), (6,4), (2,2). Which of these locations should be selected to minimize the cost of flow movement and why? Assuming rectilinear distances.
  - *Evaluating cost:* minimize total cost over time of flow
  - Option 1: cost = $440/hour
  - Option 2: cost= $470/hour
  - Option 3: cost= $550/hour

Rank the Performance: Example

- Evaluating possible exchanges in CRAFT:



$$\min z = \sum_{i=1}^{m} \sum_{j=1}^{m} f_{ij} c_{ij} d_{ij}$$

Weighted Factors

Often times there are multiple factors/criteria that are important but deal with very different aspects of the layout design:

- **Layout characteristics:**
  - Cost of movement/ distance traveled
  - Shop floor visibility
  - Aesthetics of the layout
  - Ease of expanding in the future
- **Unit load implied:**
  - Impact on WIP inventory levels
  - Space requirements
  - Impact on material handling equipment
- **Storage strategies:**
  - Space and people requirements
  - Impact on material handling equipment
  - Human factors risks
- **Material handling:**
  - Use of current equipment
  - Investment on new equipment
  - Space and people requirements

Weighted Factors

- How do you account for the relative importance of these different factors when making a decision?
  - Weighted factors
  - Prioritization matrices & Analytical Hierarchy Process

Weighted Factors

- Note: A weight is assigned to each factor we identify
- Then each alternative (A, B, C, …) is rated & scored
- The score is simply the rating x the weight for that factor

WEIGHTED FACTOR COMPARISON FORM

Company _____ Prepared by _____ Date _____
Facility Plan for _____
Sheet _____ Of _____

Alternatives

		A		B		C		D		E	
Factor	Weight	Rt.	Sc.	Rt.	Sc.	Rt.	Sc.	Rt.	Sc.	Rt.	Sc.
1.											
2.											
3.											
4.											
5.											
6.											
7.											
8.											
9.											
10.											
11.											
12.											
13.											
14.											
15.											
16.											
17.											
18.											
19.											
20.											
Totals											

Figure 11.4 Weighted factor comparison form.

Weighted Factors: Example

- Four layout alternatives are to be evaluated. Five factors are to be used in evaluating the layout alternatives.
- First, we will assign a numerical weight to each factor. These can be both quantitative and qualitative and require careful consideration.

1.	Annual material handling cost	25
2.	Construction cost	15
3.	Ease of expansion	20
4.	Employee's preference	30
5.	Rail siding access	10

Weighted Factors: Example

- Each layout is then ranked on the basis of each factor, producing the following results.
- Note: It is crucial that we keep the same scale (e.g. 1-10) for each factor! These should also follow the same polarity.

	Factor	A	B	C	D
1.	Annual material handling cost	10	7	5	2
2.	Construction cost	8	4	10	9
3.	Ease of expansion	7	8	10	9
4.	Employee's preference	6	8	7	10
5.	Rail siding access	10	6	5	7

Weighted Factors: Example

- Lastly, we calculate the scores and total the scores to end up with a finalized **weighted factor comparison form**

Factor	Wt.	A		B		C		D	
		Rt	Sc	Rt	Sc	Rt	Sc	Rt	Sc
1. Annual material handling cost	25	10	250	7	175	5	125	2	50
2. Construction cost	15	8	120	4	60	10	150	9	135
3. Ease of expansion	20	7	140	8	160	10	200	9	180
4. Employee's preference	30	6	180	8	160	7	210	10	300
5. Rail siding access	10	10	100	6	240	5	50	7	70
Totals	100		790		695		735		735

In-class Activity

You are given the following weighted factor comparison form for a mall that is in planning. Fill in the missing items (elements in the table).

Factor	Wt.	1		2		3		4	
		Rt	Sc	Rt	Sc	Rt	Sc	Rt	Sc
1. Aesthetic design	20	4	80	10	200	**A**	140	2	40
2. Construction cost	10	9	90	4	**B**	8	80	6	60
3. Space & people requirements	15	10	150	7	105	8	120	10	150
4. Ease of expanding	**C**	6	30	3	15	5	25	9	45
Totals	50		**D**		360		365		295

In-class Activity

Solution

Factor	Wt.	1		2		3		4	
		Rt	Sc	Rt	Sc	Rt	Sc	Rt	Sc
1. Aesthetic design	20	4	80	10	200	**7**	140	2	40
2. Construction cost	10	9	90	4	**40**	8	80	6	60
3. Space & people requirements	15	10	150	7	105	8	120	10	150
4. Ease of expanding	**5**	6	30	3	15	5	25	9	45
Totals	50		**350**		360		365		295

Analytical Hierarchy Process

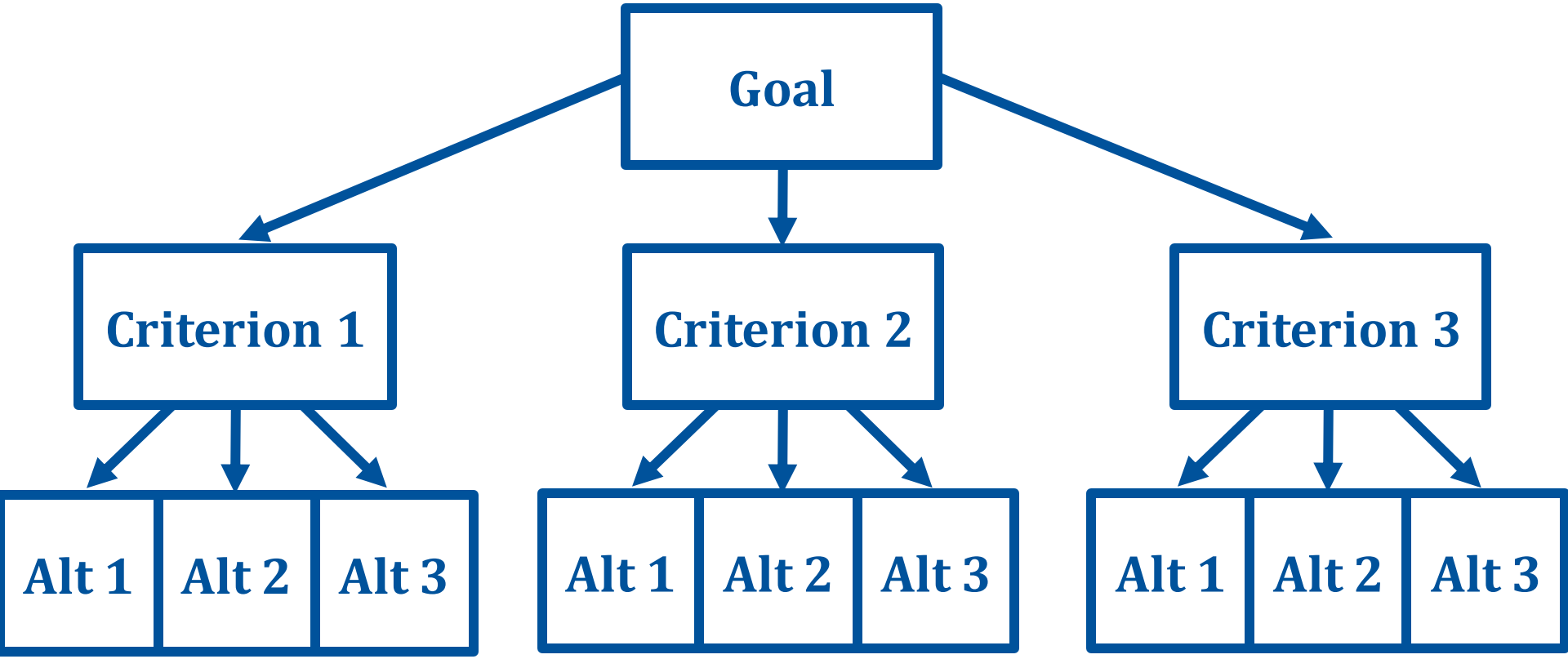
The **Analytical Hierarchy Process (AHP)** - Not in the book!

- Deals with complex decision making
- Sets priorities
- Reduces complex decisions to a series of pairwise comparisons
- Captures both subjective and objective aspects of a decision

Implementation of the AHP:

- Computing the vector of **criteria weights**
- Computing the matrix of **option/alternative scores**
- Ranking the options

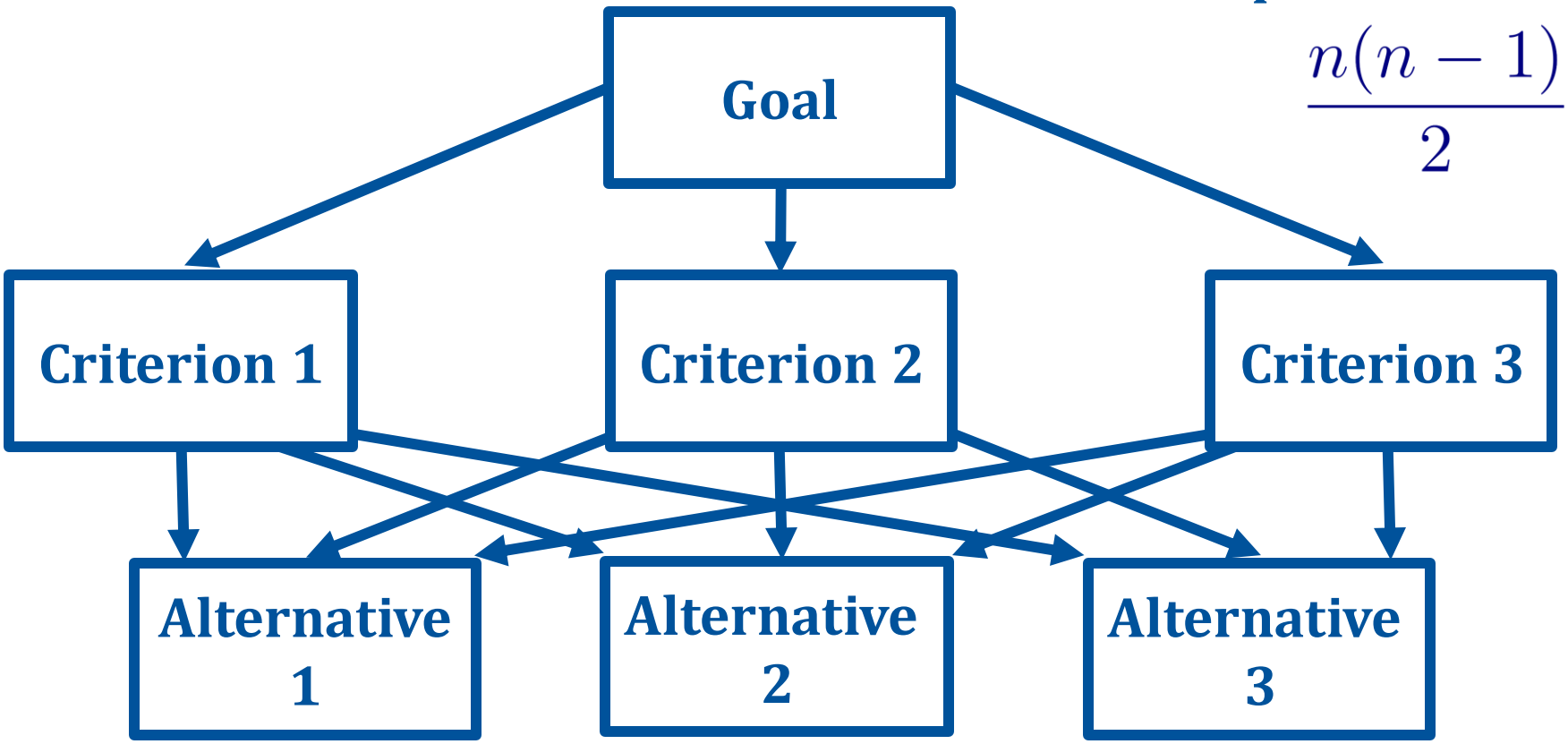
Analytical Hierarchy Process



Analytical Hierarchy Process

Note # of pairwise comparisons:

$$\frac{n(n-1)}{2}$$



Analytical Hierarchy Process

- Create a pairwise comparison matrix **A**, which is n x n, and n is the number of alternatives
- Each entry $\mathbf{a_{jk}}$ of the matrix A represents the importance of the $\mathbf{j^{th}}$ alt relative to the $\mathbf{k^{th}}$ alt
- The relative importance between two criteria is measured using a numerical scale (usually 1-9)
- Compute the normalized comparison matrix $\mathbf{A_{norm}}$ from A
- The criterion weight vector **w** (m-dimensional column vector) is calculated by averaging the entries on each row of A_{norm}

$$\bar{a}_{jk} = \frac{a_{jk}}{\sum_l^m a_{lk}}$$

a_{jk}	Interpretation
1	j and k are equally important
3	j is slightly more important than k
5	j is more important than k
7	j is strongly more important than k
9	j is absolutely more important than k

AHP – Example 1

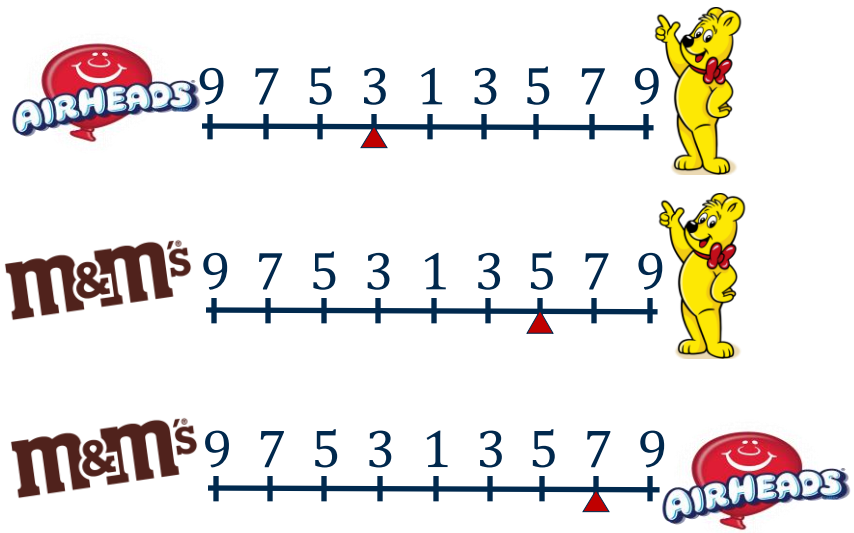
Ex1: There are three choices of candy in a movie theater:

How do we go about picking one?

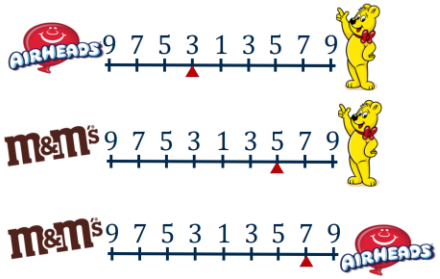
1. First, make **pairwise comparisons**

In this case we are just comparing the yumminess factor!

In this case we only have 3x(3-1)/2=3 comparisons



AHP – Example 1



Ex: There are three choices of candy in a movie theater:

2. Make the comparison matrix **A**

- If the judgement value is on the side of k, we put the actual judgement value
- If the judgement value is on the side of j, we put the reciprocal of the judgement value
- Note: The values on the bottom side of the diagonal are simply reciprocals of the top side

$$\mathbf{A} = \begin{array}{c|ccc} & & k & \\ & H & A & M \\ \hline j\ H & 1 & 1/3 & 5 \\ A & 3 & 1 & 7 \\ M & 1/5 & 1/7 & 1 \end{array}$$

AHP – Example 1

Note: w can also be used to evaluate criterions!

Ex: There are three choices of candy in a movie theater:

3. Make the priority matrix **w**

- **Sum** each columns of the **A** matrix

$$\mathbf{A}= \begin{array}{c|ccc} & H & A & M \\ \hline H & 1 & 1/3 & 5 \\ A & 3 & 1 & 7 \\ M & 1/5 & 1/7 & 1 \\ \hline \text{Sum} & 21/5 & 31/21 & 13 \end{array}$$

- **Divide** each element of the matrix with the sum of its column to make a normalized relative weight

$$\mathbf{A}_{\text{norm}}= \begin{array}{c|ccc} & H & A & M \\ \hline H & 5/21 & 7/31 & 5/13 \\ A & 15/21 & 21/31 & 7/13 \\ M & 1/21 & 3/31 & 1/13 \\ \hline \text{Sum} & 1 & 1 & 1 \end{array}$$

AHP – Example 1

$$\mathbf{A}_{\text{norm}} = \begin{array}{c|ccc} j \backslash k & H & A & M \\ \hline H & \frac{5}{21} & \frac{7}{31} & \frac{5}{13} \\ A & \frac{15}{21} & \frac{21}{31} & \frac{7}{13} \\ M & \frac{1}{21} & \frac{3}{31} & \frac{1}{13} \\ \text{Som} & 1 & 1 & 1 \end{array}$$

Ex: There are three choices of candy in a movie theater:

3. Make the priority matrix **w**

- **Average** across the rows to get a **vector w**

$$w = \begin{array}{c|ccc} j \backslash k & H & A & M \\ \hline H & \frac{5}{21} + \frac{7}{31} + \frac{5}{13} \\ A & \frac{15}{21} + \frac{21}{31} + \frac{7}{13} \\ M & \frac{1}{21} + \frac{3}{31} + \frac{1}{13} \end{array} \times \frac{1}{3} = \begin{bmatrix} 0.28 \\ 0.64 \\ 0.074 \end{bmatrix}$$

- The priority vector **w** shows the relative weights among the alternatives we want to compare
  - Haribo: 28.28%
  - **Airheads: 64.34% (we will select airheads)**
  - M&M's: 7.38%

Note: all elements in **w** add up to 1 (100%)

AHP – Example 2

Ex2: There are three choices for a late night binge TV show, pick the best one based on the scale below (which compares enjoyment):

AHP – Example 2

Ex2: There are three choices for a late-night binge TV show, pick the best one based on the scale below (which compares enjoyment):

$$\begin{array}{c|ccc} & NG & F & HIMYM \\ \hline NG & 1 & 5 & 9 \\ F & 1/5 & 1 & 7 \\ HIMYM & 1/9 & 1/7 & 1 \\ \hline Sum & 59/45 & 43/7 & 17 \end{array}$$

$$\begin{array}{c|ccc} & NG & F & HIMYM \\ \hline NG & 45/59 & 35/43 & 9/17 \\ F & 9/59 & 7/43 & 7/17 \\ HIMYM & 5/59 & 1/43 & 1/17 \\ \hline Sum & 1 & 1 & 1 \end{array}$$

$$\omega = \begin{array}{c} \\ NG \\ F \\ HIMYM \end{array} \begin{array}{c} \begin{array}{ccc} NG & F & HIMYM \end{array} \\ \begin{bmatrix} 45/59 + 35/43 + 9/17 \\ 9/59 + 7/43 + 7/17 \\ 5/59 + 1/43 + 1/17 \end{bmatrix} \end{array} \times \frac{1}{3} = \begin{bmatrix} 0.70 \\ 0.24 \\ 0.06 \end{bmatrix}$$

In this case, we end up picking **New Girl!**

AHP – Multiple Criteria

Now consider when there is **more than one criteria** for each alternative

Let's say we want to compare **n** *different alternatives* using **m** *criterions*

We want to build the matrix of **alternatives' scores** that is a **n x m** matrix **S**. Where each entry $\mathbf{s_{ij}}$ represents the score of the $\mathbf{i^{th}}$ alternative with respect to the $\mathbf{j^{th}}$ criterion

AHP – Multiple Criteria

To build this **S** matrix:

1. Build a pairwise comparison for each of the **m** criteria, j=1…m
   - Each matrix is a **n x n** matrix (like in the previous examples), where n is the number of alternatives to be evaluated
   - Each entry of the matrix represents the evaluation of the **i**th alt compared to the **h**th alt with respect to the **j**th criterion
2. Calculate the **normalized matrix** and the **score** vector **s** *(same process as what we did in the examples)* for the alternatives of each of the **m** criteria
3. Finally, we obtain the option score matrix **S** by combining the score vectors (**s**) for each **m** criteria

S=[s(1)…s(m)]

AHP – Multiple Criteria

Ranking the options:

4. Once the weight vector **w** (from prioritization) and the score matrix **S** have been computed, we can compute a vector **v** of the final weighted scores by multiplying S and w (matrix multiplication)

$$c_{11}=a_{11}b_{11}+a_{12}b_{21}+a_{13}b_{31}+a_{14}b_{41}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \\ b_{41} & b_{42} & b_{43} \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \end{bmatrix}$$

2 x 4 4 x 3 2 x 3

v=S•w

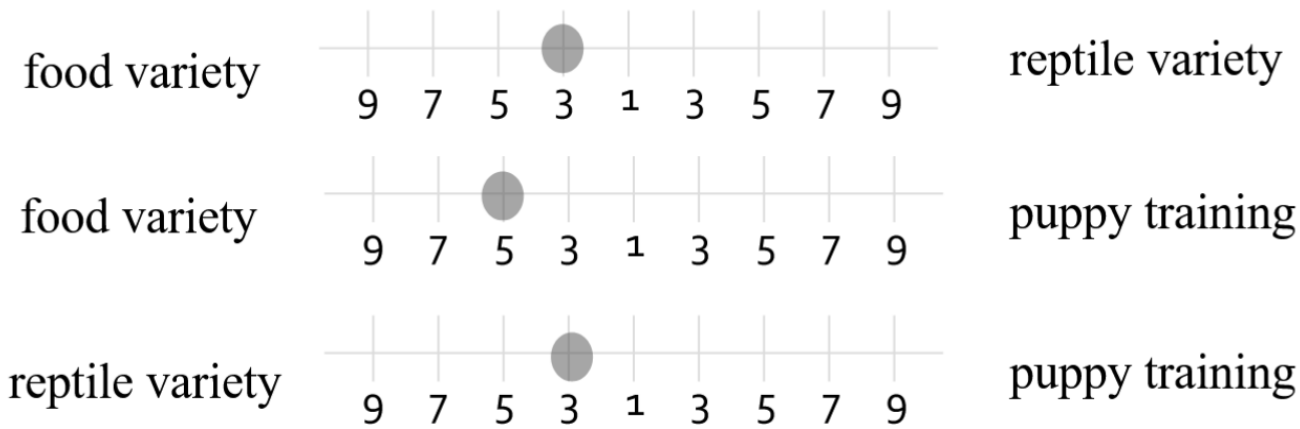
Note: in this case the priority matrix **w** is computed using criterions!

AHP – Multiple Criteria Example

Let's consider three pet stores that we want to rank based on three different criteria:

1. food variety
2. reptile variety
3. puppy training

Pairwise comparison of criteria



AHP – Multiple Criteria Example

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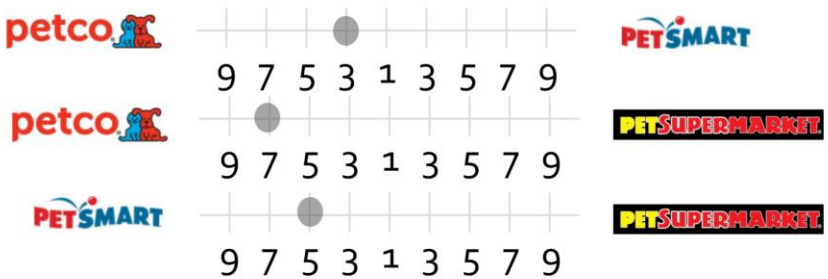
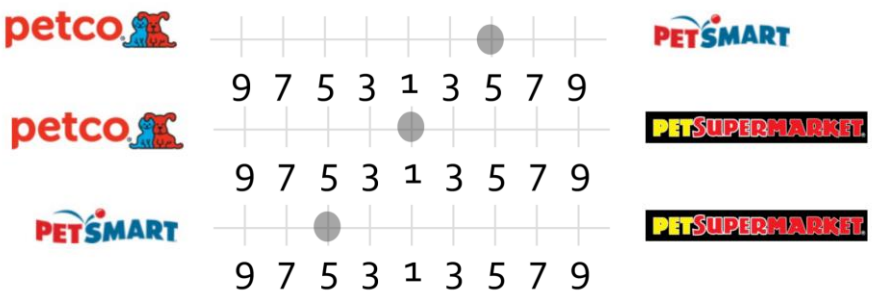
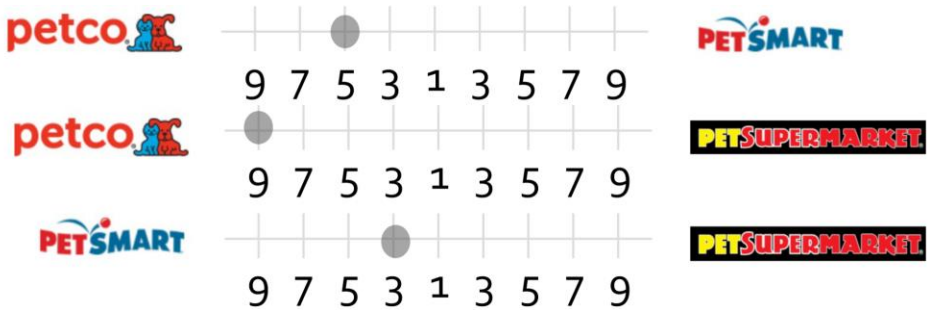
1.



2.



3.



AHP – Multiple Criteria Example

First, we will rank the pet stores across each one of our criteria: 1) food variety, 2) reptile variety, 3) puppy training

$$\begin{array}{c} \\ \text{petco} \\ \text{PETSMART} \\ \text{PETSUPERMARKET} \end{array} \mathbf{B}^{(1)} = \begin{array}{c} \begin{array}{ccc} \text{petco} & \text{PETSMART} & \text{PETSUPERMARKET} \end{array} \\ \begin{bmatrix} 1 & 3 & 7 \\ 1/3 & 1 & 5 \\ 1/7 & 1/5 & 1 \end{bmatrix} \end{array}, \quad \mathbf{B}^{(2)} = \begin{array}{c} \begin{array}{ccc} \text{petco} & \text{PETSMART} & \text{PETSUPERMARKET} \end{array} \\ \begin{bmatrix} 1 & 1/5 & 1 \\ 5 & 1 & 5 \\ 1 & 1/5 & 1 \end{bmatrix} \end{array}, \quad \mathbf{B}^{(3)} = \begin{array}{c} \begin{array}{ccc} \text{petco} & \text{PETSMART} & \text{PETSUPERMARKET} \end{array} \\ \begin{bmatrix} 1 & 5 & 9 \\ 1/5 & 1 & 3 \\ 1/9 & 1/3 & 1 \end{bmatrix} \end{array}$$

AHP – Multiple Criteria Example

The corresponding **score vectors s** for each of the criteria:

food variety

$$s^{(1)} = [\ 0.643\ 0.283\ 0.074\ ]^T,$$

petco

reptile variety

$$s^{(2)} = [\ 0.143\ 0.714\ 0.143\ ]^T,$$

PETSMART

puppy training

$$s^{(3)} = [\ 0.748\ 0.180\ 0.072\ ]^T.$$

petco

The **score matrix S** then ends up looking like this:

$$\mathbf{S} = [s^{(1)} \quad s^{(2)} \quad s^{(3)}] = \begin{bmatrix} 0.643 & 0.143 & 0.748 \\ 0.283 & 0.714 & 0.180 \\ 0.074 & 0.143 & 0.072 \end{bmatrix}$$

AHP – Multiple Criteria Example

Now we will compute the prioritization score **w** from our criteria

$$\mathbf{S} = [s^{(1)} \quad s^{(2)} \quad s^{(3)}] = \begin{bmatrix} 0.643 & 0.143 & 0.748 \\ 0.283 & 0.714 & 0.180 \\ 0.074 & 0.143 & 0.072 \end{bmatrix}$$

$$\mathbf{w} = \left\{ \begin{matrix} 0.633 \\ 0.260 \\ 0.106 \end{matrix} \right\} \begin{matrix} \text{Food variety} \\ \text{Reptile variety} \\ \text{Puppy training} \end{matrix}$$

Finaly, we can calculate the final weighted scores vector **v** (note there are some previous rounding errors)

$$\boldsymbol{v} = \mathbf{S} \cdot \boldsymbol{w} = [\boxed{0.523} \; 0.385 \; 0.092]^T.$$

petco PETSMART PETSUPERMARKET

Here, we ended up picking **Petco**, even when it did not have the best reptile variety

AHP – Dealing with real values

What do we do when dealing with real, actual values? Like gas mileage, cost, time

Instead of ranking these, just use the actual values to create the comparison, **A**, matrix

For example:

- Two restaurants: health score
  - R1: 89
  - R2: 95

A	R1	R2
R1	89/89=1	89/95=0.94
R2	95/89=1.07	95/95=1

AHP – Dealing with real values

Be careful that the A matrix reflects the true decision made!

If we have two options, X & Y, then the largest of the two should represent the **"good"** decision

So larger values in the A matrix denote that one alt is better than the other

AHP – Dealing with real values

For example:

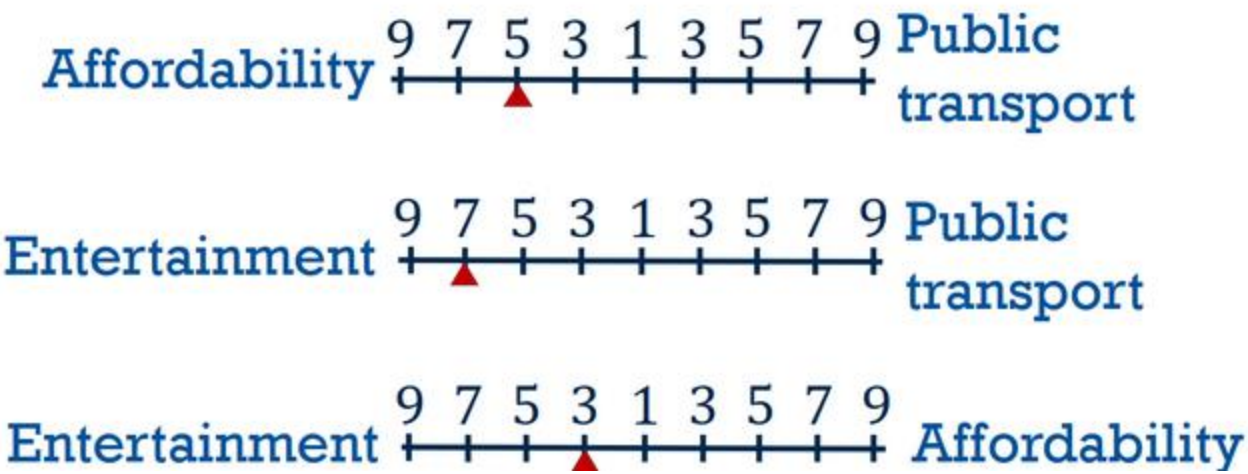
- Say X & Y are two routes used for commuting
- Our data is provided in minutes
    - X: 14 mins
    - Y: 11 mins

In this case we would prefer the shortest route. So how can we transform these values to construct our A matrix?

Simple solution: Take the reciprocals, this way we can reward the shortest commute times

In-class Activity

You and your friends are picking a travel destination for the holidays. There are 3 options: NYC, LA, and Seattle, and 3 criteria: affordability, public transport, and entertainment. You are given the **s** vectors for each alternative, and the following ranking for the criteria. Which destination should we pick?



NYC, LA, Seattle

$\mathbf{s}_{aff}=[0.2, 0.3, 0.5]^T$

$\mathbf{s}_{pbt}=[0.7, 0.1, 0.2]^T$

$\mathbf{s}_{ent}=[0.3, 0.6, 0.1]^T$

In-class Activity

Solution

$$\begin{array}{c|ccc} & \text{Aff} & \text{PbT} & \text{Ent} \\ \hline \text{Aff} & 1 & 9 & 1/3 \\ \text{PbT} & 1/5 & 1 & 1/7 \\ \text{Ent} & 3 & 7 & 1 \\ \hline & \frac{21}{5} & 13 & \frac{31}{21} \end{array} \xrightarrow{\text{norm}} \begin{array}{c|ccc} & \text{Aff} & \text{PbT} & \text{Ent} \\ \hline \text{Aff} & 5/21 & 5/13 & 7/31 \\ \text{PbT} & 1/21 & 1/13 & 3/31 \\ \text{Ent} & 15/21 & 7/13 & 21/31 \end{array} \xrightarrow{\text{avg. rows}} \begin{array}{c|c} 0.283 & \text{Aff} \\ 0.074 & \text{PbT} \\ 0.643 & \text{Ent} \end{array} = \omega$$

$$S = [S_{Aff}, S_{PbT}, S_{Ent}] = \begin{array}{ccc|c} 0.2 & 0.7 & 0.3 & \text{NYC} \\ 0.3 & 0.1 & 0.6 & \text{LA} \\ 0.5 & 0.2 & 0.1 & \text{Seattle} \end{array}$$

$$v = S \cdot \omega = \begin{array}{c|c} 0.301 & \text{NYC} \\ 0.478 & \text{LA} \\ 0.221 & \text{Seattle} \end{array} \rightarrow \text{pick LA}$$